volume 8, issue 2, May 2006

identification news

on the move

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PHILIPS

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Gearing up for kick off

Football fans around the world are counting down to June 9 and the start of the 2006 FIFA World Cup™. 32 of the world's best national teams and over 3 million fans will come together for a festival of skill and passion. And for the sixth consecutive tournament, Philips is an Official Partner of the FIFA World Cup™, enhancing people's enjoyment of the beautiful game's biggest event.

The great ones make it look simple

At its best, football looks effortlessly simple. The same is true for technology. As an Official Partner, we're providing advanced yet easy-to-use solutions that bring people closer to the action and atmosphere, whether they're in the stadium, at home or congregating with friends and fellow fans (*see page 5*).

Those lucky enough to have a ticket will benefit from quicker access to the stadium thanks to a specially developed electronic ticketing infrastructure installed at each of the 12 host stadiums. Four stadiums holding FIFA World Cup™ football matches use Philips' MIFARE contactless infrastructure – Cologne, Munich, Frankfurt and Hannover. As a leader in contactless smartcard technologies, we were the natural choice to offer our expertise and know-how for this infrastructure. Philips smart card chip technology has already proven successful for the ticketing at the FIFA Confederations Cup in Germany last year.

Smooth access to the action

We made recommendations to the organizing committe (OC) for defining the system specifications to meet the needs of the organizers and fans. Our contactless smart card technology should provide smooth access to the stadium and protection against fraud.

Fully compliant with the ISO 14443A standard and our MIFARE contactless platform, the final specification has already been approved and released by the committee. The host stadiums are in the final stages of rolling out and testing the system.

Tickets with a brain

Each of the 3.2 million tickets contains a MIFARE contactless smart card IC with a unique ticket number. To enter the stadium, fans just wave their ticket over the readers at the turnstile and walk through. Addressing the concerns of the German data protection organization, no personal details are stored on the chip.

The system includes a number of security measures to prevent fraud and ensure as many real fans as possible can enjoy the game. Firstly, the ticket holder's name and address are printed on the ticket when it is issued. Police will carry out random spot checks to compare the personal details on the ticket with the holder's passport. This should stop ticket touts buying up large numbers of tickets and selling them on at much higher prices.

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FIFA WORLD CUP GERMANY 2006
OFFICIAL PARTNER

In addition, the unique code stored on each ticket's chip makes it impossible to gain entry with a counterfeit ticket. If necessary, issued codes can be blocked so stolen tickets cannot be used.

A homogenous system

"Using Philips technology has eased the complexities linked to ticketing such as counterfeiting and the misallocation of tickets to individuals banned from the tournament. The technology ensures even if tickets are lost or stolen before the event, they can be cancelled and re-issued to the approved owner. Smart card technology is something global sporting events should be considering to optimize fan services – a key objective in staging the 2006 FIFA World Cup™," said Horst R. Schmidt, Senior Vice-President OC 2006 FIFA World Cup™.

With this deployment, Germany has become the first country with a common sports infrastructure – an infrastructure that will continue to offer sports fans many benefits even after the FIFA World Cup™ is over. For instance, host clubs will be able to offer faster access to their own fans and to season ticket holders from visiting clubs. And with many more stadiums, both in Germany and other countries, already expressing an interest in the solution, these benefits will rapidly become more widespread.

Looking further to the future, the infrastructure is easily extendible with new applications. So fans may soon be using their match tickets to travel to and from the game or even pay for merchandise or refreshments within the stadium.

For more information on Philips and the 2006 FIFA World Cup™: www.philips.com/fifaworldcup

For more information on MIFARE:
www.philips.semiconductors.com/products/identification/mifare/

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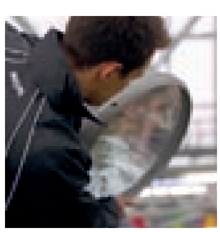

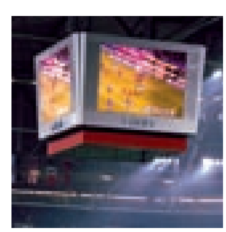

Playing to win

As our feature on the 2006 FIFA World Cup™ illustrates, sports events are gaining real benefits from contactless smart card infrastructures. But it's not just football stadiums that use this technology: supporters and competitors in all kinds of sports are also benefiting from secure contactless solutions.

For example, the FLEX-Tag from Swiss systems integrator Microtronic AG has been a great success at sporting and other events. Using MIFARE technology, it first appeared at the 2004 Olympic Games as a Coca Cola bottle-shaped key fob. More recently, it was used at the 2006 Winter Olympics in Turin where athletes and officials used it to get free 'POWERADE' drinks from 200-plus MIFARE-equipped vending machines.

Basketball and beyond

MIFARE technology also enabled a smart card project for the 2005 European Basketball Championship in Serbia and Montenegro. Installed by SDD Information Technology Group (ITG) in Vrsac's 'Center Millennium' sports arena (which hosted one of the quarterfinal groups), the multi-application system puts the in-built functionality and versatility of MIFARE smart cards to good use.

"Philips is a leader in the identification market", stated Mr. Spira Matic, General Manager of SDD ITG. "So it only made sense to choose them as our technology partner and to use their MIFARE products, which are the de facto standard for contactless smart card systems."

Around 4500 MIFARE standard 1 K cards have been issued. Season tickets allow visitors to access the center on multiple occasions and hold information on discounts and other benefits such as membership of the separate fitness centre and sauna. VIP cards offer the same services as season tickets, but may include different benefits and are printed in a different color. Employee ID cards allow staff to access the center. All the cards can also be used as an electronic wallet, allowing people to pay for services available at the center.

Enjoying the game

From providing athletes with easy access to drinks, to giving fans and staff secure and convenient access to stadiums, contactless smart card solutions bring many benefits to the sports world. They make our lives easier and open up new possibilities for providing additional services, so that watching or participating in sports becomes an even more enjoyable experience.

For more information on MIFARE:
www.philips.semiconductors.com/products/identification/mifare/

Philips at the 2006 FIFA World Cup™

The smart card chip technology is just one of the ways we're helping fans and organizers enjoy the 2006 FIFA World Cup™.

We are also:

- lighting 8 of the 12 stadiums enabling a great view and some great action
- creating giant screens for public viewing in all 12 host cities so everyone can revel in the atmosphere
- equipping stadiums with Vidiwalls to keep fans on top of the action
- providing stadiums and teams with easy-to-use HeartStart defibrillators in case of emergency
- enabling viewers around the world to experience the entire event in high-definition and widescreen (16:9)
- supplying ten thousand FlatTV™ sets across the 12 stadiums for media, broadcasting and official use

Saving money, saving lives

Healthcare touches everyone's lives, dealing literally with life and death issues. However, it is also a large industry, which is facing major challenges worldwide. Not least of these is cost. For example, the average healthcare spending across European Union countries is almost 9% of the Gross Domestic Product (GDP)[1]. And with public expectation of the medical profession increasing and the world's population aging, the healthcare industry urgently needs solutions that address these rising costs.

What's more, despite increasingly large investment and the best efforts of healthcare professionals, mistakes still happen. In the United States alone, 98,000 people each year die because of medical errors[2]. That's more than the country's death toll from road accidents, breast cancer or HIV/AIDS. Meanwhile, errors cost the United Kingdom's National Health Service (NHS) some £500 million (USD 875 million)[3] annually.

As a Healthcare, Lifestyle and Technology company, we want technology to improve people's lives. We believe that applying RFID and contactless technologies to the healthcare sector has huge potential to improve patient care and save lives. Not only can these technologies reduce dangerous and expensive mistakes, they can dramatically improve efficiency in numerous areas. This will let healthcare professionals focus more time on patients, and will free up more money for patient care and research rather than administration.

Beating the drug counterfeiters

The pharmaceutical industry will likely be the first healthcare sector to reap the benefits of RFID. Globally, the pharmaceutical market is worth around USD 500 billion[4]. Unfortunately, this huge market is a tempting target for the unscrupulous. It's estimated that around 5% of drugs administered globally are counterfeit, rising to 80% in some individual countries[5].

Counterfeit drugs do huge damage to a brand's reputation, affecting the manufacturer's standing and ability to fund research into new medicines. Even more worryingly, they can be extremely dangerous. In China alone, some 180,000 people a year die as a result of taking fake medicines[6].

RFID offers a way to secure the pharmaceutical supply chain. If drugs are RFID tagged at the pallet, case and

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package level, they can be quickly and accurately tracked from the manufacturer to the pharmacy. As they enter and leave each link in the distribution chain (manufacturer, distributor, wholesaler and retailer), the drugs are scanned and authenticated. Any counterfeits can be immediately identified and removed from circulation before they do harm.

In addition to the safety and brand protection benefits from deterring counterfeiters, RFID can also improve operational efficiency throughout the pharmaceutical supply chain. This will help ensure a consistent supply of medicines while reducing costly overstocking and return of out-of-date drugs. Currently, drug returns cost the industry USD 2 billion annually[7]. >

1 Source: EFPIA
2 Source: US Food and Drug Administration (FDA)
3 Source: NHS
4 Projection based on 2002 value and growth rate. Source: IMS Health
5 Source: RECONNAISSANCE International
6 Source: World Health Organization
7 Source: HDMA presentation at Auto-ID Healthcare Adoption Forum

Industry initiatives

Many industry and regulatory bodies are already actively promoting the use of RFID technology in the pharmaceutical industry. In the US, the world's largest healthcare market, the Healthcare Distribution Management Association (HDMA) wants Electronic Product Code (EPC) tagging at case level by 2006 and item level by 2007. Meanwhile, the US Food and Drug Administration (FDA) is calling for pallet, case and package tagging plus the installation of RFID infrastructure at manufacturers, wholesalers, hospitals and most retailers by 2007.

"As a Healthcare, Lifestyle and Technology company, we want technology to improve people's lives".

Pharmaceutical companies are also investigating RFID. One of the first to really put it to the test is Pfizer who recently launched a major program to tag all Viagra® sold in the USA. Pharmacists and wholesalers can now authenticate their stocks using specially designed scanners to read the unique EPC incorporated into each pallet, case and package.

"Drug counterfeiting is growing," explains Tom McPhilips, vice president of Pfizer's US Trade Group. "Our aim is to enhance patient safety. We are creating additional barriers for criminals attempting to counterfeit our products. Patients who use our medicines should feel confident they're getting the authentic product, not potentially dangerous fakes."

Streamlining hospital care

RFID and contactless technologies also have huge potential for healthcare within hospitals. Hospitals are complex entities. They have a lot of portable equipment that needs tracking for regular maintenance. They also use huge amounts of drugs and medical supplies. RFID could streamline and simplify the huge asset, inventory and supply-chain management challenges within hospitals, greatly reducing operational overheads.

However, some of the most exciting benefits come from clinical applications of these technologies. One important example is in the testing of blood samples. Testing labs have invested heavily in automation and now have huge analytical capacity. But there is one major bottleneck. Logging and pre-analytical processing still has to be done by hand, which is error prone and leads to backlogs and delays.

A British trial is using Philips RFID technology to tag test tubes for samples. The aim is to reduce testing errors, cut >

The hospital, tomorrow

From May 16 – 19 at the Paris Expo, The hospital, tomorrow exhibition will give the public a glimpse into the future of healthcare. Philips is a key sponsor of the event, and we're also showcasing a number of advanced healthcare solutions.

A MIFARE-based ID card helps identify patients anywhere within the hospital. It stores administrative information like the patient's name, address and health service number. It also records an overview of all treatments, providing an accurate log that can be easily and securely updated.

The medication tracking system ensures patients receive the right drug in the right dose and at the right time. A mobile Near Field Communication (NFC) device contains a complete delivery schedule for medications. When a delivery time is prompted, the nurse is guided to the patient where she authenticates the patient's ID, confirms she has the right drug and logs when she administered the medication all via NFC technology.

The remote blood pressure monitor allows doctors to keep a close eye on patients in their own homes. The patient takes their own blood pressure with an easy-to-use measuring device. Then they just touch the measuring device to their cell phone to transfer the data using NFC technology, which is then sent over the cell phone network to their doctor or hospital.

operating costs and improve staff efficiency both in the lab and on the ward. This would ultimately result in faster, more accurate and more patient-centered care.

This kind of solution has big market potential. For example, the UK alone tests more than 150 million samples per year. And globally, testing volume is expected to double over the next five years.

The missing link

Clearly RFID offers huge benefits for the logistical challenges faced by hospitals. However, these benefits can only be truly realized if contactless technology is also applied to one more critical area: patient identification. That's the view of Paul Schmidt, founder of Proxximity, the company overseeing the UK trial. As a full-time hospital doctor, Paul comes across these issues every day. *(See page 12 for more on Proxximity)*

"Hospitals are increasingly becoming integrated electronic environments with computer systems sharing information between departments," Paul explains. "There's just one gap: the patient. Replacing existing paper wristbands with electronic identification will let doctors close the loop. Then we can focus more on patient care and less on administration, and help ensure patients get the right tests, treatment and medication."

Complete integration

For the biggest patient benefits, RFID / contactless technologies must be implemented throughout the healthcare sector. That means tagging drugs, medical supplies, samples and patients. Doctors and nurses could access a patients' complete medical history at the bedside simply by scanning their wristbands. So they know straightaway they're administering the right treatment to the right patient.

Such integrated solutions are still some way off. But as a leader in RFID and contactless technologies, we're working with partners across the value chain to put all the pieces in place. We're delivering advanced and highly reliable solutions that improve efficiency, protect brands and increase patient safety. Solutions that save money as well as lives.

For more information on identification products:
www.semiconductors.philips.com/products/identification

For more information on NFC:
www.semiconductors.philips.com/products/identification/nfc

Partnership boosts ICODE infrastructure

More than ever before, close cooperation between companies is important in attaining common business goals. The world of RFID is no exception and Philips works hard to forge strong relationships both within and related to the industry. Our partnership with leading manufacturer FEIG ELECTRONIC is a prime example – sharing ideas and development has led to the introduction of a next generation ICODE reader. >

The success of Philips ICODE products in the market depends substantially on how easily a reader infrastructure can be established. Our partners in building that infrastructure are key, since only a handful of companies have the capability to build the required infrastructure and system solutions. In that respect Feig is a valued partner.

Philips and Feig have been working together successfully for more than 15 years and on many different projects. "The collaboration that resulted in the new smart label card reader began some 18 months ago," said Peter Schmallegger, Philips' Product Marketing Manager for ICODE. "The goal was to develop a new HF reader that offered superior performance and supported the latest market trends and regulations".

"Sharing ideas and inspiration really helps drive development"

As with all our partnerships, each member brings their own expertise to the project. For the reader project both companies contributed to product development, with Feig's capability as a reader manufacturer complemented by Philips' in-depth knowledge of RFID. "Working together not only makes maximum use of resources, it also allows efforts to be coordinated to ensure the end-solution is aligned with everyone's requirements," said Frithjof Walk, Feig's Sales & Marketing Manager for OBID®. "Our new reader is a good example – we can offer customers an improved infrastructure, at the same time satisfying new smart label technologies from Philips."

Available already, the new smart label reader represents the next generation in reader infrastructure. By going back to the drawing board, we were able to make significant improvements to all aspects of performance including reading distance and speed. It incorporates new technology that can be exploited by the latest smart labels, such as Philips ICODE SLI-S tags with password protection *(see article on page 14)*.

A key focus when developing the reader was to make it closer to a 'straight-out-of-the-box' solution. So along with enhanced performance a number of special features make it much easier to set up and use. These include a new antenna tuning concept that makes tuning less critical, a standard interface compatible with Feig's UHF readers, and an improved user-interface. Overall, it is very flexible and easier to integrate into a system.

"Partnering with Philips has proven very successful," said Frithjof. "Sharing ideas and inspiration really helps drive development, and we can rely on each other if there are problems to overcome. Certainly we faced challenges together, but we've learned a lot from each other and had some fun too."

For more information on ICODE:
www.philips.semiconductors.com/products/identification/icode

Courtesy: FEIG ELECTRONIC

"Thankfully, Philips also understands the importance of patient ID."

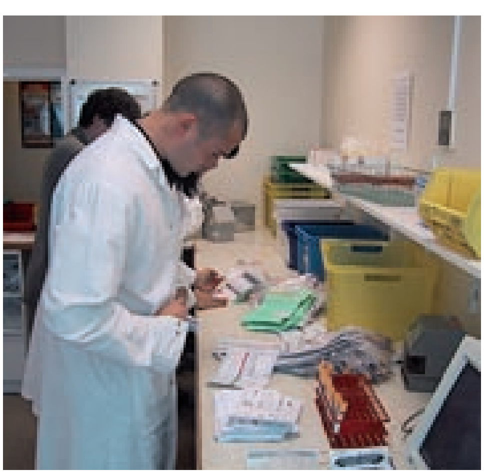
Partner Perspective

In this edition of Partner Perspective we talk to **Proxximity** founder and CEO, Dr. Paul Schmidt. Paul tells us about this small company's big ideas for improving efficiency and patient care in our hospitals through the intelligent application of RFID technology.

Paul Schmidt, founder and CEO

"Healthcare is probably one of the last industries still to go through an efficiency leap," Paul begins. "Doctors have to document everything they do and end up doing everything twice: once to actually do it and once to document it."

Not only does this mean doctors spend more time on administration and less with patients, it causes delays and mistakes that could endanger the patient and waste money.



Courtesy: Proxximity

Proxximity

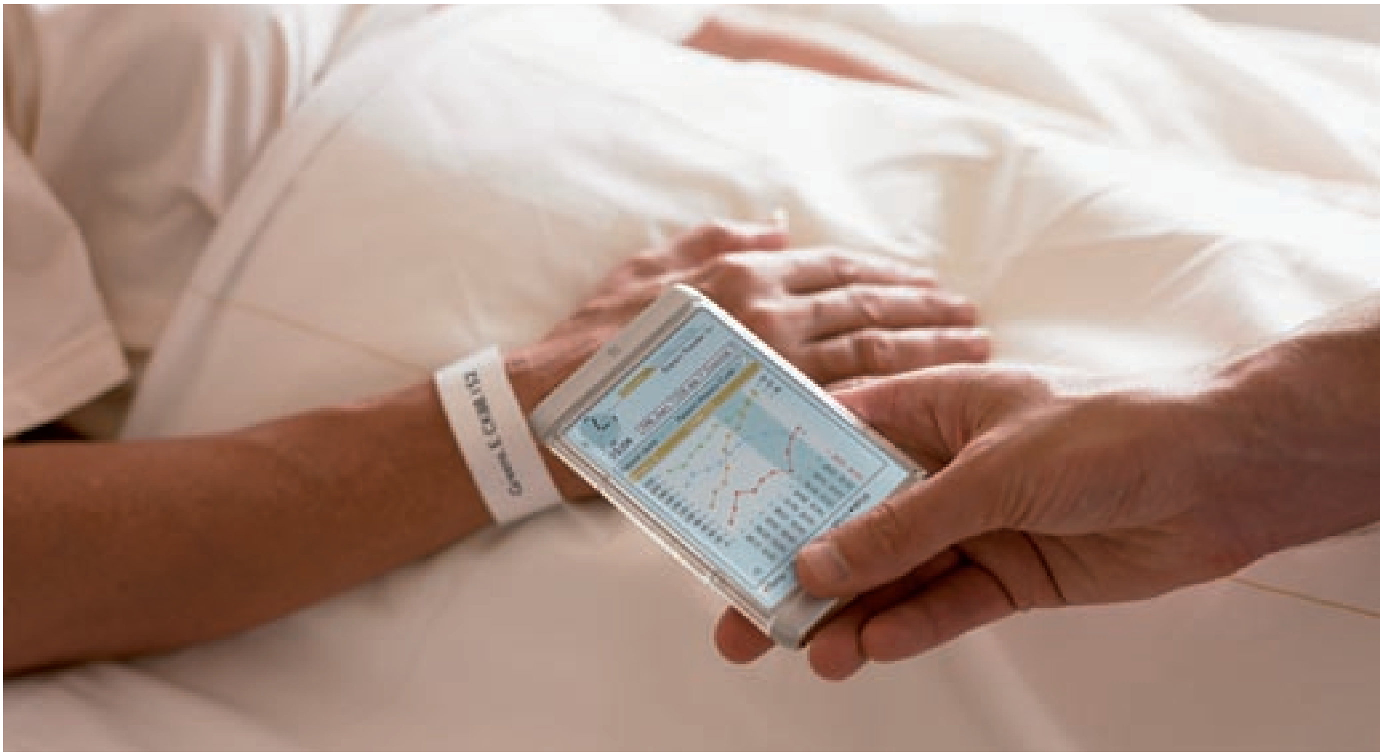
That's why Paul got together with a fellow doctor to start Proxximity in 2001. With their intimate knowledge of the sharp end of healthcare, their goal was to find a better way to create an automated audit trail – a system that would allow doctors to record their actions as they work at the bedside and update patient records as they carry out treatment.

"We had the business perspective," Paul points out. "But we needed to find out how technology could help."

Building partnerships

Hospitals already have quite extensive IT networks. However, they don't reach to the bedside, which is precisely where doctors need to obtain and transfer information. As Paul puts it, "We have to make the patient part of the hospital's electronic environment." That means electronically securing the patient's identity and matching it to their medical records.

Proxximity began looking at RFID and soon realized that creating their own proprietary solution would not be feasible. "We needed to find partners and initiate collaborations between bigger companies who were already working in this area. That's difficult for a small company to do, but Philips has the weight and credibility to bring all the necessary players together," said Paul. >



"Thankfully, Philips also understands the importance of patient ID," Paul continues. "It's not the largest-volume RFID application, but it's vital for improving healthcare efficiency. And it opens the door to other higher-volume uses like tagging samples for testing."

Putting it to the test

Building on these partnerships, Proxximity has set up a case study to test the RFID tagging of blood samples. Pathology laboratories have expensive automated testing equipment, but the initial manual blood sample checking is slow and labor intensive, preventing laboratories from fully realizing the benefits of their investment in equipment.

In the trial, patients wear an RFID wristband that looks just like the traditional paper wristband. When samples are taken, they are put into RFID-tagged test tubes and the patient's details are transferred to the tube. The pathology lab's computer reads the data from the tube to find out exactly which patient the sample belongs to and what tests are required.

The work isn't yet complete but the early signs are encouraging. The case study predicts RFID systems will cut pre-analytical processing time by 90%, guarantee 100% data accuracy and reduce other errors by 80%. "A conservative calculation suggests the system could save the UK's National Health Service £50 million (USD 90 million) per annum."

The last word

Laboratory managers want a system that eliminates errors at their source. Doctors need a system that is fast, intuitive and reliable. RFID offers an elegant solution to both these demands. The test tube tags can carry all necessary information. This ensures only the correct samples are taken, reducing waste and inconvenience to patients. It also allows RFID systems to work reliably in healthcare settings even when there is no network link to the lab.

"We set out to create a system that would allow healthcare professionals to spend more time with their patients, that would save money and improve patient safety," Paul concludes. "Our role has evolved, and now we're more of a consultant and enabler to the value chain. But the potential benefits to patients are so big, we would be happy just to see a solution out there."

For more information on ICODE:
www.philips.semiconductors.com/products/identification/icode

For more information on MIFARE:
www.philips.semiconductors.com/products/identification/mifare

Take security to the next level

The Philips ICODE SLI-S IC offers an enhanced security option to the smart label market – password protection. It lets you tailor the security level to meet your own needs. For example, you can prevent unauthorized access to sections of the on-chip memory or allow read-only access while only authorized parties can modify the data stored.

Similarly, the password protection prevents people from illicitly switching off the EAS anti-theft functionality. In addition, the ICODE SLI-S IC supports customer requirements such as kill and privacy commands. To meet growing application needs, it features 2 Kbits of on-board memory with a 64-bit unique serial number and 96-bit, one-time programmable electronic product code (EPC).

The ICODE SLI-S is fully compliant with ISO 15693, ISO 18000 and EPC protocol standards as well as the widespread ICODE SLI and ICODE EPC infrastructure. Therefore it can be easily integrated into existing projects and is supported by numerous hardware partners. It offers reading speeds up to 200 labels/sec using the EPC inventory command set and 60 labels/sec with the fast inventory read command set.

ICODE SLI-S

- Optional password-protected memory access
- 2048 bits of on-chip memory
  - 64 bits UID
  - 96 bits EPC (OTP)
  - 1280 bits programmable user memory
- Interface complies with ISO 15693, ISO 18000-3 and EPCglobal HF Class 1 standard
- Multi-label operation up to 60 tags/sec according to the ISO protocol and up to 200 tags/sec according to the EPC protocol
- Reading distances up to 2 m
- Runs on the same infrastructure as ICODE SLI, ICODE 1, ICODE EPC and ICODE UID
- 23 pF and 97 pF (for smaller labels) versions
- Available in volume from Q3 2006

Key applications

- Item-level tagging in:
  - libraries
  - rental services
  - the retail sector
  - pharmaceutical supply chains
  - the fashion industry

For more information on ICODE:
http://www.philips.semiconductors.com/products/identification/icode/

Identification updates

First commercial NFC rollout

The 'Hanauer Handy Ticket' application ('Farschein Hanau') is the world's first commercial deployment of NFC-enabled ticketing. From April, people in Hanau, Germany can use a cell phone in place of a ticket when traveling on local buses. Following on from a ten month trial in cooperation with Philips, Rhein-Main-Verkehrsverbund (RMV) and Nokia, Vodafone is now selling Nokia 3220 cell phones equipped with NFC-enabled ticketing at its Hanau shops. Vodafone customers can also enjoy special offers with the 'RMV ErlebnisCard Hanau' at 14 selected vendors in and around the city.

Courtesy: Rhein-Main-Verkehrsverbund

Using NFC-enabled ticketing is very straightforward. When boarding or alighting from the bus, the passenger simply passes the phone over the reader. The entry or exit is confirmed on the phone's display while the travel details are saved in the phone, allowing the user to check information such as the most recent journeys. At the end of each month, the customer receives a bill including a list of all journeys taken.

In addition to the transport application, the RMV ErlebnisCard Hanau application turns the phone into a bonus card, giving customers additional benefits relating to purchases or when visiting an event. The cell phone then serves as an NFC key – with access rights saved, verified and charged via the phone.

Vodafone is already looking to use NFC for other applications, and to integrate secure-chip functionality into SIM cards. This will allow customers to retain their data when they switch phones. While this is the first commercial NFC rollout, you can expect to see the technology being used in all kinds of contactless ticketing and payment applications in the near future.

For more information on Philips' NFC mobile solutions: www.philips.semiconductors.com/applications/smart_cards/nfc_mobile/

NFC Forum grows

The NFC Forum has recently accepted several new principal- and associate-level members, increasing its membership to over 70 organizations. Its newest principal member is South Korean cellular network operator SK Telecom. Since the Forum's inception in 2004, global interest in NFC has grown steadily. The Forum works to advance and set standards for NFC, the contactless payment and information-downloading technology. This year, as many as 18 NFC pilots are expected to take place in Asia, Europe and the United States.

For more information on the Forum: www.nfc-forum.org/home

Meet us at the following events

Event: **Smart Card Conference**
Date: 30 May - 1 June, 2006
Location: Beijing, China
Website: www.smartcards-china.com

Event: **ICAO Interoperability testing session**
Date: 30 May - 2 June, 2006
Location: Berlin, Germany
Website: www.wg8.de/interoptest-berlin

Event: **Electronic Passport Forum**
Date: 8 - 9 June, 2006
Location: Paris, France
Website: www.electronic-passport.com

Event: **IFA**
Date: 1 - 6 September, 2006
Location: Berlin, Germany
Website: www.messe-berlin.de

Event: **Card Expo India**
Date: 6 - 8 September, 2006
Location: New Delhi, India
Website: www.electronicstoday.org/smartcardsexpo.htm

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